Time (UTC or Mission Elapsed Time)	Robot Function / Mode	Primary Objective / Milestone	Operation Mode (Teleop / Autonomous / Semi-Autonomous / Scripted)	Inputs (e.g., command, signal, state)	Expected Outputs (e.g., telemetry, sensor data)	Data Products Generated	Verification Method	Associated Risk or Sensitivity (High/Med/Low)
Launch day, 0-1 (Mission Elapsed Time)	Launch And Commissioning	Spacecraft Separation and initial health checks	Scripted/Autonomous	Launch vehicle separation signal, onboard startup sequences	Spacecraft status telemetry, power system telemetry, communication link confirmation	Initial spacecraft health report, subsystem status logs	Verify successful separation communication establishment, and nominal subsystem health/	High
Cruise Phase, Years 0-6	Cruise And Approach	Maintain spacecraft health and perform trajectory correction maneuvers	Semi-Autonomous	Ground-planned trajectory updates, star tracker measurements, navigation state estimates	Navigation telemetry, spacecraft health telemetry, maneuver confirmation	Trajectory correction logs, navigation solution history	Verify trajectory remains within mission tolerance and all spacecraft systems remain operational	Medium
Europa Orbital Insertion and Reconnaissance	Europa Orbit Insertion And Orbital Reconnaissance	Deploy orbital agents and identify suitable landing region	Autonomous	Orbital state estimates, terrain imaging data, radar survey measurements	Surface maps, hazard maps, candidate landing sites	Global terrain map, landing site assessment report	Verify successful orbital insertion, orbiter deployment, and identification of viable landing zones.	High
Surface Operations Day 0-5	Landing, Surface Checkout, And Site Characterization	Deploy rover, verify system health, and calibrate science instruments.	Autonomous	Landing state, rover deployment commands, sensor calibration routines	Health telemetry, localization estimates, calibration measurements	System checkout report, calibration datasets, local terrain map	Verify successful rover deployment, communications, localization accuracy, and payload calibration	High
Surface Operations Day 5-35	Surface Survey And Ice Characterization	Map subsurface ice structure and identify candidate cryobot insertion sites	Autonomous	Ground-penetrating radar returns, orbital terrain map, localization state estimations	Ice thickness estimates, terrain classifications, rover position estimates	Subsurface radar maps, ice thickness profiles, candidate cryobot site database, consensus terrain maps	Verify radar measurements meet required accuracy and candidate sites satisfy mission selection criteria	Medium
Surface Operations Day 35-50	Data Transmission And End Of Mission	Downlink mission data, select final cryobot site, and safely passivate system	Semi-Autonomous	Stored science data, final site selection results, mission completion commands	Mission archive transmission, final health telemetry, passivation confirmation	Final mission report, validated cryobot insertion site package, complete science archive	Verify all science data successfully transmitted and spacecraft enters safe passivated state	Low

Dependencies / Predecessor Actions	Estimated Duration	Notes
Successful vehicle loading and spacecraft deployment	1 day	
Completion of launch and commision phase	~6 years	
Successful arrival at Europa and orbital insertion burn	30 Days	
Reconnaissance complete and landing site selected	5 Days	
Successful rover deployment and payload calibration	30 Days	
Completion of science objectives and final site selection	15 days	